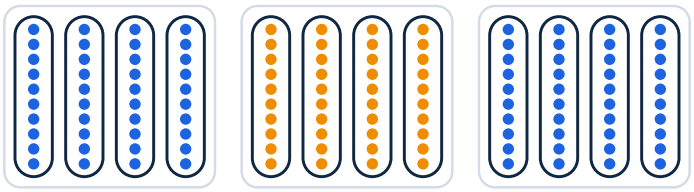
Multiplying by Tens

Groups of tens: use the fact you know, then count tens.

3 groups of 4 tens



$3 \times 40 = 120$

3 groups of 4 tens = 12 tens
12 tens is 120.

Use a fact you know

$6 \times 7 = 42$

$6 \times 70 = 42 \text{ tens} = 420$

Same fact, but now it counts tens.

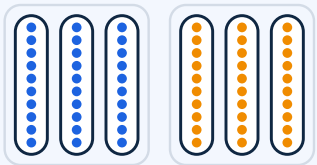
Careful with 5s

$5 \times 4 = 20$

$5 \times 40 = 20 \text{ tens} = 200$

The fact already ends in 0.
20 tens is 200: two zeros.

Your turn!



2 groups of 3 tens = 6 tens

$2 \times 30 =$ 60

For teachers

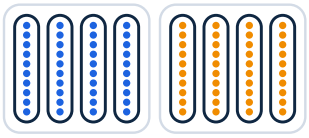
Goal: multiply a one-digit number by a multiple of 10 (3.NBT.3). Say "6 × 70 is 6 groups of 7 tens. 6 × 7 = 42, so it is 42 tens, 420." Avoid "just add a zero": 5 × 40 = 20 tens = 200 shows why.

Multiplying by Tens: Practice

Use the fact you know. Count the tens.

Part 1 Groups of tens Count the groups and the tens in each group.

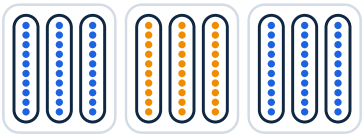
1)



2 groups of 4 tens = 8 tens

2 × 40 = 80

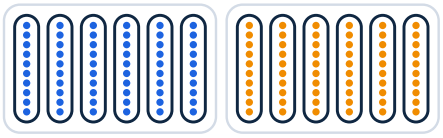
2)



3 groups of 3 tens = 9 tens

3 × 30 = 90

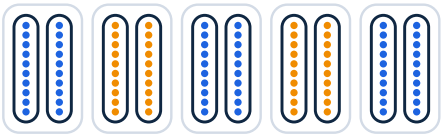
3)



2 groups of 6 tens = 12 tens

2 × 60 = 120

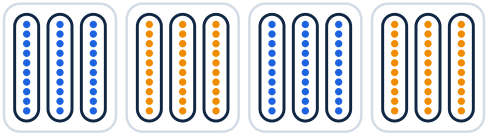
4)



5 groups of 2 tens = 10 tens

5 × 20 = 100

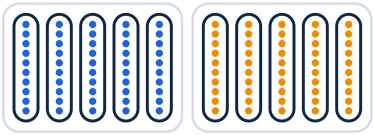
5)



4 groups of 3 tens = 12 tens

4 × 30 = 120

6)



2 groups of 5 tens = 10 tens

2 × 50 = 100

Part 2 How many tens? Multiply, then write how many tens.

7) 3 × 5 tens = 15 tens

8) 4 × 4 tens = 16 tens

9) 6 × 2 tens = 12 tens

10) 7 × 3 tens = 21 tens

11) 2 × 8 tens = 16 tens

12) 5 × 3 tens = 15 tens

Multiplying by Tens: Practice (continued)

Part 2 How many tens? (continued)

13) 8 × 2 tens = 16 tens

14) 3 × 7 tens = 21 tens

15) 4 × 6 tens = 24 tens

Part 3 The fact, then the tens The first answer helps with the second.

16) 6 × 4 = 24 so 6 × 40 = 240

17) 3 × 8 = 24 so 3 × 80 = 240

18) 7 × 5 = 35 so 7 × 50 = 350

19) 4 × 9 = 36 so 4 × 90 = 360

20) 9 × 3 = 27 so 9 × 30 = 270

21) 5 × 6 = 30 so 5 × 60 = 300

22) 8 × 7 = 56 so 8 × 70 = 560

23) 6 × 8 = 48 so 6 × 80 = 480

Part 4 Times 2 and times 5 No help this time.

24) 2 × 70 = 140

25) 5 × 80 = 400

26) 9 × 20 = 180

27) 5 × 50 = 250

28) 2 × 90 = 180

29) 7 × 20 = 140

30) 5 × 90 = 450

31) 4 × 50 = 200

32) 6 × 50 = 300

Part 5 Harder facts No help this time. Find the fact first.

33) 6 × 70 = 420

34) 8 × 40 = 320

35) 7 × 90 = 630

36) 9 × 60 = 540

37) 3 × 60 = 180

38) 8 × 80 = 640

Multiplying by Tens: Practice (continued)

Part 5 Harder facts (continued)

39) $4 \times 70 = 280$

40) $9 \times 90 = 810$

41) $7 \times 60 = 420$

42) $6 \times 90 = 540$

43) $3 \times 90 = 270$

44) $8 \times 30 = 240$

Part 6 Tens first No help this time. 30 × 4 is the same as 4 × 30.

45) $30 \times 4 = 120$

46) $60 \times 3 = 180$

47) $70 \times 8 = 560$

48) $90 \times 4 = 360$

49) $80 \times 5 = 400$

50) $20 \times 3 = 60$

51) $50 \times 7 = 350$

52) $40 \times 8 = 320$

53) $60 \times 6 = 360$

Part 7 Mixed No help this time.

54) $7 \times 70 = 490$

55) $4 \times 80 = 320$

56) $9 \times 40 = 360$

57) $2 \times 20 = 40$

58) $8 \times 90 = 720$

59) $3 \times 40 = 120$

60) $6 \times 30 = 180$

61) $9 \times 70 = 630$

62) $5 \times 70 = 350$

63) $8 \times 50 = 400$

64) $7 \times 40 = 280$

65) $3 \times 20 = 60$

Part 8 Remember? Problems from earlier packets.

66) 7 groups of 6 = 42

67) Round 64 to the nearest ten: 60

Multiplying by Tens: Practice (continued)

Part 8 Remember? (continued)

68) half of 18 = 9

69) 703 − 245 = 458

70) 348 + 275 = 623

71) 4 groups of 8 = 32

Part 9 Word problems How many groups? How many in each?

72) A box holds 50 crayons. How many crayons are in 3 boxes?

Number sentence 3 × 50 = 150

Answer 150 crayons

Draw or show your work

73) Each train car holds 70 people. How many people fit in 3 cars?

Number sentence 3 × 70 = 210

Answer 210 people

Draw or show your work

74) A game gives 20 points for each star. Ben gets 6 stars. How many points is that?

Number sentence 6 × 20 = 120

Answer 120 points

Draw or show your work

75) A farmer plants 4 rows with 60 seeds in each row. How many seeds is that?

Number sentence 4 × 60 = 240

Answer 240 seeds

Draw or show your work

Multiplying by Tens: Practice (continued)

Part 9 Word problems (continued)

76) Challenge: A store has 9 boxes of 50 pencils. It sells 75 pencils. How many pencils are left?

Number sentence $9 \times 50 = 450, 450 - 75 = 375$

Answer 375 pencils

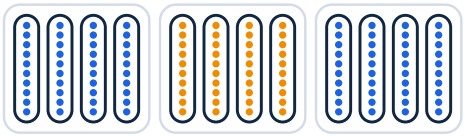
Draw or show your work

Check-In: Multiplying by Tens

Show what you know. No help this time.

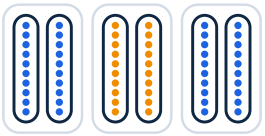
Mastery: 8 of 10 (80%)

Part 1 Groups of tens

1) 

3 groups of 4 tens = 12 tens

3 × 40 = 120

2) 

3 groups of 2 tens = 6 tens

3 × 20 = 60

Part 2 The fact, then the tens

3) 2 × 6 = 12 so 2 × 60 = 120

4) 9 × 5 = 45 so 9 × 50 = 450

Part 3 Multiply

5) 8 × 70 = 560

6) 6 × 60 = 360

7) 4 × 60 = 240

8) 7 × 50 = 350

9) 6 × 40 = 240

10) 8 × 20 = 160